TOPIC 02

Correlation vs Regression

Two ways of looking at the same pair of variables: how closely they move together, and how one affects the other

WHAT YOU WILL LEARN

- State what correlation measures and what regression models
- List the differences in direction, number of outputs and graph
- Convert between the slope b_1 and the correlation r
- Explain why the regression of y on x differs from x on y
- Say when each tool is the right one

NOTE Where this connects

The previous topic showed that **correlation does not imply causation**: films and GDP moved together without affecting each other. Regression is built for the opposite job: it states how a change in one variable (x) is associated with a change in another (y), and it is only meaningful when that direction makes sense. This topic puts the two tools side by side using the SAT and college-GPA data from the next topic.

1. Two tools, two questions

	Correlation	Regression
Question answered	How strongly do x and y move together?	How does y change when x changes? What y do we expect for a given x?
Output	One number r between −1 and +1	An **equation** $\hat{y} = b_0 + b_1x$, plus fit measures
Direction	**Symmetric**: r(x, y) = r(y, x)	**One-way**: y depends on x (the roles cannot be swapped freely)
Causality	Does not capture it; only the degree of relationship	Built on an assumed cause-and-effect direction
Units	Unit-free	b_1 is in units of y per unit of x
Graph	A **single point** summarising the cloud (r)	The **best-fitting line** through the data points, minimising the distances to them
Use	Spotting and measuring association	Prediction and explanation

The lecture's example: in the Vin Diesel data the variables **move together** but do not cause each other, so a correlation can describe them while a regression of GDP on films would be nonsense. With income and education, a regression makes sense: education affects income, though not the other way within the same period.

KEY IDEA Symmetric versus one-way

Correlation between income and education equals correlation between education and income. Regression of income **on** education is a different model from regression of education **on** income.

2. The same data, seen both ways

We use 84 students: x is the SAT score and y is the college GPA at graduation (the data set from the lecture workbook). The correlation and the regression line come from the same numbers.

Quantity	SAT (x)	GPA (y)
Mean	1845.3	3.330
Standard deviation	104.53	0.2716
Correlation r	0.6372	

The correlation is the **same whichever variable is called x**. The regression slope is not.

R^2 (explained in Topic 04) is the share of the variability in y that the line explains. In simple regression it is r^2.

3. How r and the slope are linked

$$b_1 = r\frac{s_y}{s_x} \qquad b_0 = \bar{y} - b_1\bar{x}$$

The slope is the correlation rescaled by the ratio of standard deviations. Check with our data: b_1 = 0.6372 × 0.2716/104.53 = **0.001656**, the same as the least-squares slope 0.001656.

- If r = 0, then b_1 = 0: the best line is flat.
- The sign of b_1 is the sign of r.
- In **simple** regression, R^2 (the share of variability explained, Topic 04) is exactly r^2. Here r^2 = 0.4060 and R^2 = 0.4060.

4. Why there are two different regression lines

Take the same scatter plot and regress GPA on SAT (the usual direction), then SAT on GPA (the reverse). The slopes are not reciprocals of each other, so the two lines are different.

Regression	Equation	Slope meaning
GPA on SAT (y on x)	GPA = 0.275 + 0.00166 × SAT	0.00166 GPA points per SAT point
SAT on GPA (x on y)	SAT = 1028.6 + 245.2 × GPA	245.2 SAT points per GPA point

If the two were the same line, the second slope would be 1/0.00166 = 604. It is 245. Only when the points lie **exactly** on a line (|r| = 1) do the two lines coincide. The weaker the correlation, the further the two lines spread apart; they always cross at the point of the two means.

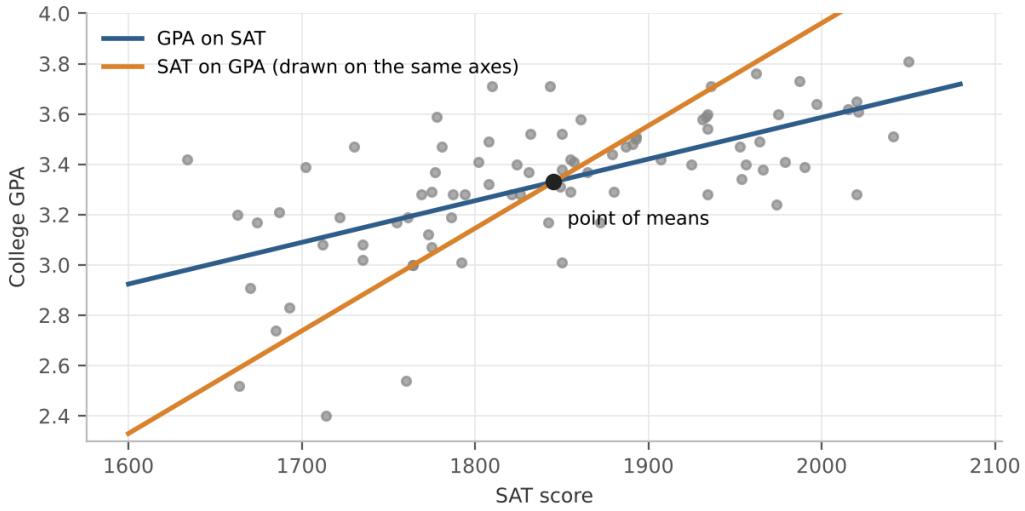


Figure 1. The two regression lines for the same data. They cross at the means and differ because the correlation is well below 1.

A four-point check. Take x = 1, 2, 3, 4 and y = 2, 1, 4, 3. Then r = 0.6 and both standard deviations are equal, so both slopes equal r × 1 = 0.6. The line of y on x has slope 0.6; the line of x on y, written with y on the horizontal axis, also has slope 0.6, but on the usual axes that second line has slope 1/0.6 = 1.67. They are different lines. A tidy fact: **the product of the two slopes equals r^2**. For the SAT data, 0.001656 × 245.22 = 0.4060, which is R^2.

> **COMMON MISTAKE** **Which one makes sense?**
> Students take the SAT first and earn their GPA four years later, so the sensible model predicts GPA from SAT. The reverse regression asks the later outcome to 'cause' the earlier score. The next topic returns to this point.

5. Choosing between them

If you want to…	Use
Describe how strongly two variables are associated, with no direction	Correlation
Predict one variable from another, or explain it	Regression
Quantify the change in y for a one-unit change in x	Regression (slope b_1)
Check whether a plausible causal story holds	Neither alone; study design and logic are needed

6. In Python

```
import numpy as np
from scipy import stats

data = np.loadtxt("satgpa.csv", delimiter=",", skiprows=1)
sat, gpa = data[:, 0], data[:, 1]

r = np.corrcoef(sat, gpa)[0, 1]
print(round(r, 4), np.isclose(np.corrcoef(gpa, sat)[0, 1], r))   # symmetric

res = stats.linregress(sat, gpa)                          # gpa on sat
print(res.slope, res.intercept, res.rvalue ** 2)
print(r * gpa.std(ddof=1) / sat.std(ddof=1))              # same slope
```

SUMMARY

- Correlation: one symmetric, unit-free number for the strength of a linear relationship. It does not capture causality.
- Regression: an equation with a direction, used for prediction and explanation. Its line is the best fit through the points.
- $b_1 = r \times s_y/s_x$; in simple regression $R^2 = r^2$.
- Regressing y on x is not the same as regressing x on y; the lines differ unless $|r| = 1$.
- Choose the direction from logic and timing, not from the data.

Practice questions

Try each question before reading the answer underneath it.

Q1. If r(x, y) = 0.5, what is r(y, x)?

Answer. 0.5. Correlation is symmetric.

Q2. $s_x = 10$, $s_y = 4$ and $r = 0.8$. Find the slope of y on x and of x on y.

Answer. y on x: $b_1 = 0.8 \times 4/10 = 0.32$. x on y: $0.8 \times 10/4 = 2.0$.

Q3. Why can a high r not be used to claim that x causes y?

Answer. r only measures how closely the variables move together; coincidence, a confounder or reverse causation can create it (Topic 01).

Q4. Simple regression gives $r = -0.7$. What is R^2, and is the slope positive or negative?

Answer. $R^2 = 0.49$. The slope is negative, like r.

Q5. A scatter plot has $r = 1$. What do the two regression lines look like?

Answer. They are the same line, because all points lie exactly on it.